higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

SYSTEM ANALYSIS AND DESIGN
NQF LEVEL 4

1 MARCH 2019

This marking guideline consists of 12 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	B
	1.1.2	A
	1.1.3	D
	1.1.4	D
	1.1.5	A
	1.1.6	D
	1.1.7	C
	1.1.8	B
	1.1.9	C
	1.1.10	A

(10 × 1) (10)

1.2	1.2.1	Reality
	1.2.2	prototype
	1.2.3	availability
	1.2.4	checksum
	1.2.5	model
	1.2.6	observation
	1.2.7	open
	1.2.8	Consequence
	1.2.9	firewall
	1.2.10	professional

(10 × 1) (10)

1.3	1.3.1	True
	1.3.2	True
	1.3.3	False
	1.3.4	False
	1.3.5	False
	1.3.6	True
	1.3.7	True
	1.3.8	False
	1.3.9	True
	1.3.10	True

(10 × 1) (10)

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1.4	1.4.1	E
	1.4.2	A
	1.4.3	H
	1.4.4	I
	1.4.5	B
	1.4.6	J
	1.4.7	L
	1.4.8	C
	1.4.9	D
	1.4.10	M

(10 × 1) (10)
[40]

TOTAL SECTION A: 40

SECTION B

QUESTION 2

2.1 2.1.1 Yes (1)

2.1.2
- Educating people of the damaging effects of piracy may change the way they think and view piracy.
- Educating people appeals to their good nature which may make people realise piracy is wrong. (2)

2.1.3
- Use hardware dongles to protect programs.
- Manufacture software programs that first need to be registered with the organisation before they can work.
- Established organisations should be mandated to have a software policy addressing piracy.
- Require employees of organisations to sign antipiracy statements. (4)

2.1.4
- Software piracy negatively impacts software-publishing companies in terms of revenue.
- Software piracy creates unfair competition for legitimate software publishers.
- It exposes organisations to legal, financial and reputational damage. (Any 2 × 1) (2)

2.2 When an individual or organisation sells computers preloaded with illegal copies of software (1)

2.3 A set of written rules that explains how professional people working in a particular profession are allowed to behave and conduct themselves (2)

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2.4
- The Institute of Technology Professionals of South Africa (formerly CSSA)
- The Black Information Technology Forum
- Women in Tech ZA
- The Cape Innovation and Technology Initiative
- The Joburg Centre for Software Engineering
- The Information Technology Association of South Africa
- The Computing Technology Industry Association (Any 2 × 1) (2)

[14]

QUESTION 3

3.1 Availability of information refers to the assurance of reliable access to information by authorised people. (1)

3.2 Difference
- Confidentiality is the rules that limit access to information whereas integrity is the assurance that the information is accurate and valid.

Similarity
- Only authorised people can access, read or be given confidential information. Likewise, only authorised people are allowed to create, delete and make changes to confidential information. (2 + 2) (4)

3.3
- Characterise the information system.
- Identify the major risks, threats and vulnerabilities.
- Determine the likelihood of risks.
- Conduct an impact analysis.
- Determine the risks.
- Recommend controls.
- Document results. (7)

3.4 System engineering focuses on the development and organisation of complex systems over their life cycles.

OR

System engineering gives a set of procedures, strategies and techniques that a development team can apply to the development process. (1)

3.5
- Functional requirements
- Non-functional requirements
- Traceability matrix ANY (2 × 1) (2)

3.6 It refers to all the things that the project must do and achieve (scale, range, size or quantity). (2)

3.7
- Completed on time
- Meets the requirements
- Completed within the estimated budget
- Meets the required quality standard (4)

3.8 3.8.1

Checklist item	
1. Servers secured in an access-controlled room	☐
2. Access to USB ports limited	☐
3. CD drive/diskettes removed	☐
4. Hardware firewall operational	☐
5. Intrusion device operational	☐

Any other relevant answer
(Any 3 × 1) (3)

3.8.2
- Granting logon to authorised users only
- Using antivirus/antimalware software
- Regularly updating security updates from manufacturer/vendor
- Providing security locks on system cases
- Regularly auditing internal security measures (Any 4 × 1) (4)

3.9 Encryption is the process of encoding or changing a message or information in such a way that only authorised parties can access it. (1)
[29]

QUESTION 4

4.1 4.1.1
- The best way to determine whether the interviewee has understood the question is for the interviewer to listen to the answer carefully.
- The interviewer must determine whether the answer has provided the information needed.
- The interviewer must determine whether the interviewee covered the objectives that were asked in the question.
- If not, the interviewee has misunderstood the question; hence the interviewer will have to ask the question again. The interviewer must ask the question again and explain in detail so that the interviewee will understand it correctly.
- The interviewer can change the wording so that other interviewees will not make the same mistake. (Any 3 × 1) (3)

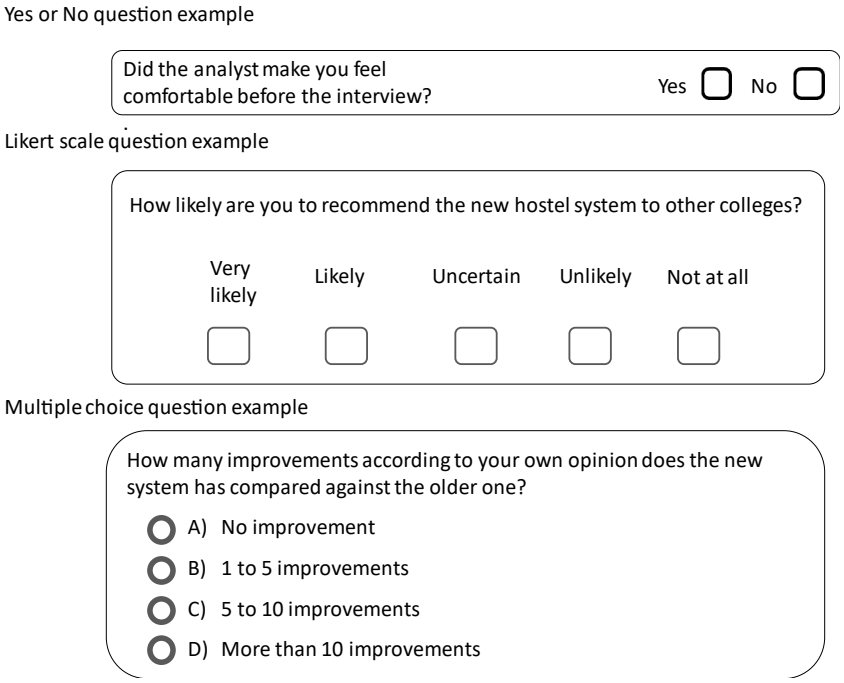
4.1.2
- Requires short, specific answers and are always limited to a set of choices, e.g. can be a true or a false and a yes or no which does not provide a lot of information and may be confusing.
- Can be a short definitive answer or direct answer to the question to provide a direct response to only that question and respondents may give simplistic responses to complex issues.
(2 × 2) (4)

4.1.3 A group interview comprises interviewing all or a few of the candidates at the same time, whereas a one-on-one interview involves only one candidate at a time. (2)

4.1.4
- Questions can be more direct and more information can be shared by the candidate.
- The interviewer can also ask additional questions to better understand the responses. Any (1 × 2) (2)

4.1.5
- What features of the system does the user use most often?
- What features does the user perform easily without any problem?
- What features cause the most frustration?
- What features does the user seldom use? (Any relevant 3 × 1) (3)

4.1.6 Example questions:



NOTE: Allocate 1 mark for the question and 1 mark for the correct indication of the type as well. (3 × 2) (6)

4.2
- Data can be gathered from users in both techniques.
- Both techniques are the easiest to solicit responses from users.
- Both techniques are cost- and time effective to gather more information.

(Any 2 × 1) (2)

4.3
- Name of data entity (E.g. table names)
- Description of data entity (E.g. Field names, data types and keys)
- Any relationship the entity is involved in

OR

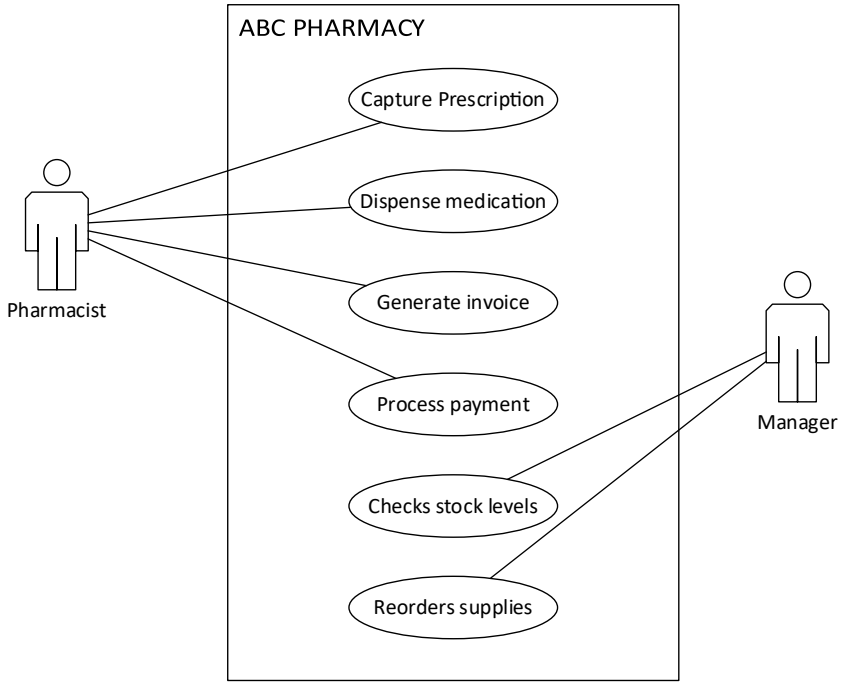
- Definitions of data flows
- Definitions of data elements
- Definitions of data stores (3)

4.4 A composite key is a combination of two or more columns/fields in a table that can be used to uniquely identify each row in the table. (2)

4.5 Preference or prejudice for or against a person or group in a way considered to be unfair (1)
[28]

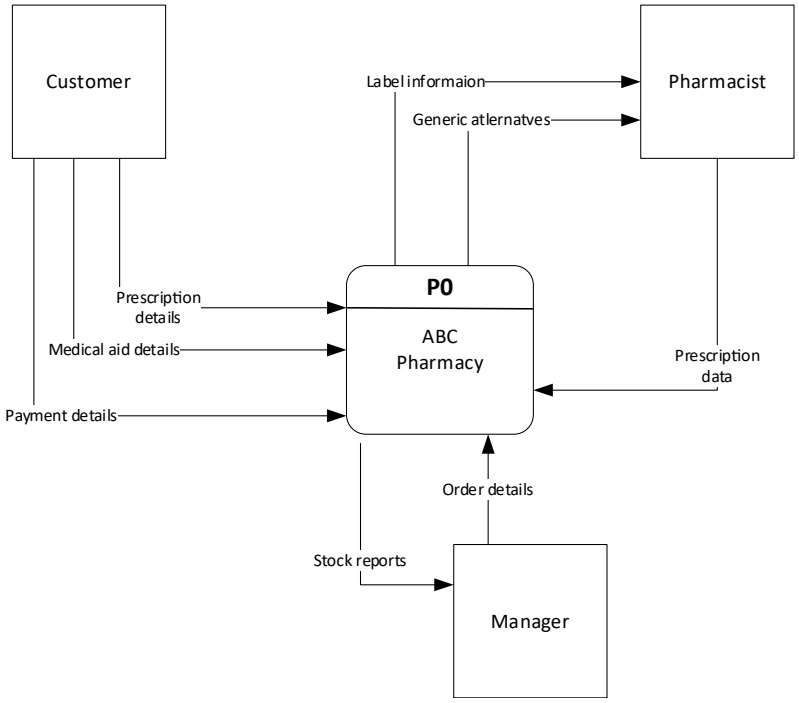
QUESTION 5

5.1



Mark allocation
1 Mark for the name of the system (1 × 1)
1 Mark for indicating the boundary (1 × 1)
1 Mark for each actor (2 × 1)
1 Mark for each applicable use case (6 × 1)
1 Mark for all the associations per actor to each use case (all must be indicated. (2 × 1) (12)

5.2



Mark allocation

1 Mark for the main process (name)	(1 × 1)
1 Mark for indicating it is a P0 or Level 0 diagram	(1 × 1)
1 Mark for each external entity	(3 × 1)
1 Mark for each data flow with direction	(8 × 1)

(13)

Deduct 2 marks if a data store is included

5.3

- Defines the application problem by conferring with clients; evaluating procedures and processes
- Develops a solution by preparing and evaluating alternative workflow solutions
- Controls the solution by establishing specifications; coordinating production with programmers
- Validates results by testing programs
- Ensures operation by training client personnel; providing support
- Provides reference by writing documentation
- Updates job knowledge by participating in educational opportunities; reading professional publications; maintaining personal networks; participating in professional organisations
- Accomplishes information systems and organisation mission by completing related results as needed (Any 3 × 1) (3)

5.4
- A flow chart details the processes to follow whereas a DFD details the flow of data through a system.
- In a flow chart, the arrows represent transfer of control (not data) between elements and the elements are instructions or decisions (or I/O, etc.).
- In a DFD, the arrows are actually data transfer between the elements, which are themselves parts of a system, whereas in a flowchart the arrows indicate the sequence of steps. (3 × 1) (3)

[31]

QUESTION 6

6.1
- Uniqueness
- Uniform size and format
- Stability
- Meaning
- Operability Any (3 × 1) (3)

6.2
- Text-based output may not be considered very attractive to users.
- Text-based output may be difficult to understand when compared to output involving graphics.
- Text-based output may not be user-friendly.
- Certain operations cannot be implemented using text-based software.

(Any 2 × 1) (2)

6.3
- Till slip / Receipt

Any relevant answer related to a Point of Sale system (1)

6.4 Online input
- Online input dialogs validate and verify data as the user enters it.
- The dialog is able to notify the user of a possible error as the user makes the mistake.
- This gives the user a chance to correct the data before they complete the form.

Off-line input
- Validation and verification is used in off-line input to ensure that the data capture has captured the information correctly.
- This applies if the capturing is either manual or automatic.
- If the system rejects a form, the respondent can resubmit the form or the system may discard the form completely. (3 + 3) (6)

[12]

QUESTION 7

7.1 7.1.1 Adaptive maintenance. (1)

7.1.2 The system is being changed or modified to be usable in the environment that is changing due to government legislation and user requests. (2)

7.1.3
- Unit testing.
- System testing.
- Integration testing. (3)

7.1.4 User-acceptance test. (1)

7.1.5 The system is tested for acceptability by the end user for compliance and requests. (1)

7.1.6 A test plan is a document that describes the scope, approach, resources and schedule of intended testing activities, whilst a test case is a formal definition of a test that defines prerequisites, an action and the expected result. (2)

7.1.7
- To check if the system has met all its objectives and satisfied all user requirements.
- To check if the system is working as efficiently as it should.
- To check if there are any lessons learnt from the design and development of this system that will benefit future systems. (3)

7.1.8
- Test plans.
- Test specifications.
- Test summary reports. (3)

7.2 7.2.1
- A new information system that runs at more than one location is referred to as a pilot-adoption approach.
- Pilot adoption is used when selecting one or two sites as pilot sites.
- Pilot sites are test sites used to demonstrate new systems.
- At pilot locations old systems will be removed and new systems installed.
- Users who still work on the old system can spend time at the pilot site and be trained how to use it.
- New systems can be tested in a live environment without completely removing the existing system.
- Once completely satisfied with the new system, the old system can be replaced at all sites. (Any 2 × 2) (4)

7.2.2
- It is an excellent way to gauge system adoption by target users.
- If problems are found in the new system, only the pilot site is affected.
- Users can be trained at the pilot sites before the new system is installed at their site.
- It eliminates large-scale risks. (Any 1 × 2) (2)

7.2.3
- A big-bang or direct adoption happens in a single instance, whereas with a pilot adoption the new system is run at a few selected pilot sites.
- With a big-bang approach the time to implement the rollout of the new system is much shorter than with a pilot adoption.
- With a big-bang approach any fall-back plans are more difficult to implement than with a pilot adoption. (Any 3 × 2) (6)

7.3 The date/time when a new information system becomes operational (1)

7.4
- The review needs to wait until bugs in the system are resolved.
- Users must be trained to use the system.
- The review should only take place once the client has used all the system functionality at least once.
- A review cannot be performed if the client has not used some parts of the system. (Any 3 × 1) (3)

[32]

QUESTION 8

8.1 An autonomous robot is a machine that performs behaviours and tasks independently and without a controller. (2)

8.2
- The inference engine being the main processing element of the expert system that applies logical rules to the knowledge base to deduce new information
- The knowledge base used to store complex structured and unstructured information used by the expert system
- The user interface by which a user interacts with the expert system by means of forms, command prompts, dialog boxes or other input methods
- Working storage that stores data specific to the problem being solved

(Any 3 × 2) (6)

8.3
- Python
- Java
- Lisp
- Prolog
- C++ (Any 3 × 1) (3)

8.4
- Household appliances
- Facial-pattern recognition
- Safety systems in automobiles
- Unmanned aerial vehicles (drones)
- Weather-forecasting systems
- Insurance-risk assessments
- Medical-diagnosis software

(Any 3 × 1) (3)
[14]

TOTAL SECTION B: 160
GRAND TOTAL: 200

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